

1. How many times does **AGGT** or its reverse complement (**ACCT**) occur in the lambda virus genome? E.g. if **AGGT** occurs 10 times and **ACCT** occurs 12 times, you should report 22.

1 point

306

2. How many times does **TTAA** or its reverse complement occur in the lambda virus genome?

1 point

Hint: **TTAA** and its reverse complement are equal, so remember not to double count.

195

3. What is the offset of the leftmost occurrence of **ACTAAGT** or its reverse complement in the Lambda virus genome? E.g. if the leftmost occurrence of **ACTAAGT** is at offset 40 (0-based) and the leftmost occurrence of its reverse complement **ACTTAGT** is at offset 29, then report 29.

1 point

26028

4. What is the offset of the leftmost occurrence of **AGTCGA** or its reverse complement in the Lambda virus genome?

1 point

450

5. As we will discuss, sometimes we would like to find *approximate* matches for P in T . That is, we want to find occurrences with one or more differences.

1 point

For Questions 5 and 6, make a new version of the **naive** function called **naive_2mm** that allows up to 2 mismatches per occurrence. Unlike for the previous questions, **do not consider the reverse complement here**. We're looking for approximate matches for P itself, not its reverse complement.



For example, **ACTTTA** occurs twice in **ACTTACTTGATAAAGT**, once at offset 0 with 2 mismatches, and once at offset 4 with 1 mismatch. So **naive_2mm('ACTTTA', 'ACTTACTTGATAAAGT')** should return the list **[0, 4]**.

Hint: See [this notebook](#)  for a few examples you can use to test your **naive_2mm** function.

How many times does **TTCAAGCC** occur in the Lambda virus genome when allowing up to 2 mismatches?

191

6. What is the offset of the leftmost occurrence of **AGGAGGTT** in the Lambda virus genome when allowing up to 2 mismatches?

1 point

49

7. Finally, download and parse the provided FASTQ file containing real DNA sequencing reads derived from a human:

1 point

https://d28rh4a8wq0iu5.cloudfront.net/ads1/data/ERR037900_1.first1000.fastq 

Note that the file has *many* reads in it and you should examine all of them together when answering this question. The reads are taken from this study:

Ajay, S. S., Parker, S. C., Abaan, H. O., Fajardo, K. V. F., & Margulies, E. H. (2011). Accurate and comprehensive sequencing of personal genomes. *Genome research*, 21(9), 1498-1505.

This dataset has something wrong with it; one of the sequencing cycles is poor quality.

Report which sequencing cycle has the problem. Remember that a sequencing cycle corresponds to a particular offset in *all* the reads. For example, if the leftmost read position seems to have a problem consistently across reads, report **0**. If the fourth position from the left has the problem, report **3**. Do whatever analysis you think is needed to identify the bad cycle. It might help to review the "Analyzing reads by position" video.